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# Moral Suspension in Human-GenAI Interaction: Scale Development and Validation for Chinese Emerging Adults

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## ABSTRACT

The emergence of generative artificial intelligence (GenAI) has introduced unprecedented ethical complexity into contemporary human-GenAI collaboration, especially among emerging adults (individuals aged 18–29). Traditional moral theories emphasize users' active rationalization processes but do not fully capture the passive ethical ambiguity that arises in interactions with opaque and autonomous GenAI systems. To address this gap, this study introduces and operationalizes the construct of Moral Suspension, defined as a psychological state characterized by interactive environmental complexity, ambiguity in responsibility attribution, and passive responsibility exemption during GenAI-assisted tasks. Guided by social cognitive theory, we developed the Moral Suspension Scale (MSS) and employed a rigorous multi-phase validation process, including expert review, pilot testing, exploratory and confirmatory factor analyses, and measurement invariance assessment. Data were collected from Chinese emerging adults ( $n = 785$ ) with practical experience in AI-assisted academic or creative activities. The finalized MSS consists of 38 items and demonstrates strong reliability, model fit, and invariance across gender. The MSS provides a robust instrument for assessing moral psychological ambiguity in GenAI contexts, advancing theoretical integration between moral psychology and human-GenAI interaction. This research contributes an empirical foundation for the ethical evaluation of GenAI and offers practical guidance for responsible system design and digital ethics education for younger generations.

## KEYWORDS

Human-GenAI interaction; moral suspension; scale development; Chinese emerging adults

## 1. Introduction

The widespread use of GenAI is changing the process of information production, knowledge acquisition, and creation. GenAI's capacity to generate human-like content, automate complex tasks, and augment human creativity is reshaping the landscape of education, work, and daily life, drawing growing scholarly and public attention to its societal value and innovative potential (Alotaibi, 2025). Despite these significant benefits, the rise of GenAI has also generated profound ethical uncertainties within human-GenAI collaboration (Farangi et al., 2024). Unlike previous automation technologies, GenAI operates via a highly autonomous probabilistic mechanism, which is characterized by output unpredictability, algorithmic opacity, and consequence attributability challenges (Arrieta et al., 2020; Chen et al., 2025); collectively, these shape a novel ethical issue. While users gain empowerment from GenAI, they confront a gray area characterized by ambiguous responsibility allocation and unknown risks (Draxler et al., 2024). Although established moral psychology frameworks like “*Moral Disengagement*” and “*Diffusion of Responsibility*” provide critical insights into how individuals rationalize misconduct within group or mediated contexts (Feng et al., 2016; A. Luo & Bussey, 2023), they emphasize active cognitive

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strategies, deliberate rationalizations to alleviate guilt. However, when faced with black-box, difficult-to-control GenAI, users frequently exhibit high reliance on its outputs (Noy & Zhang, 2023) yet tend not to attribute ownership of the content to the AI itself (Longoni et al., 2024). This contradiction results not in users actively seeking disengagement from responsibility but instead in them passively becoming entangled in a state of ambiguity regarding responsibility attribution and moral judgment.

In response, our study proposes the novel concept of “*Moral Suspension*.” This refers to an attenuation of responsibility and ethical hesitation experienced by individuals in technology (GenAI)-empowered contexts. This state arises from difficulties in clarifying the boundaries of human-machine responsibility, challenges in obtaining sufficient information to assess ethical risks, and a passive perception of algorithmic determinism. The concept aims to address the limitations of existing theories in explaining the complexities of human-GenAI interaction. It also provides a novel perspective for understanding and evaluating user ethical experiences in the GenAI era.

Drawing on Social Cognitive Theory Triadic Reciprocal Causation Model (Bandura, 1999b; Lee & Tseng, 2024), we construct Moral Suspension as a unique psychological state emerging in human-GenAI interaction, shaped by the interplay of environmental, cognitive, and behavioral ethics. The first dimension is interactive environmental complexity, which refers to the opacity and complexity of the GenAI system. GenAI’s opacity impedes users’ ability to discern the logic and sources of system-generated content, impairing their capacity to evaluate moral risks (Ross, 2024). Second is ambiguity in responsibility attribution, which refers to the uncertainty of the responsible entity in human-GenAI collaboration. Human-GenAI collaboration increasingly blurs the lines of authorship, making behavioral attribution difficult and undermining the premise of moral responsibility (Draxler et al., 2024; Weber et al., 2024). The third dimension is passive responsibility exemption, a phenomenon where technological determinism weakens personal accountability. This leads to a dynamic where users unintentionally assign blame for negative outcomes to GenAI’s limitations, as their own sense of agency is passively diminished (S. J. Bennett, 2025; Coeckelbergh, 2020; Nyholm & Smids, 2016; Smit et al., 2025; Stahl, 2006). The interplay of these dimensions forms the core experience of Moral Suspension. This psychological state challenges traditional ethical reasoning, particularly in emerging adults.

Emerging adults, aged 18–29, are critical for establishing self-identity and personal ethics (Arnett, 2007; Arnett et al., 2014). As digital natives, this population group readily adopts new technologies like GenAI, integrating it into daily life, learning, and work (Bennett et al., 2008; Coyne et al., 2013; Rau et al., 2008). GenAI’s rise coincides with emerging adults’ key moral development phase. When using GenAI for academic writing, art creation, advice seeking, or emotional support (Zhang et al., 2024), they are unprecedentedly exposed to complex new contexts of moral ambiguity. Current research mainly explores GenAI’s impact on their cognitive development, such as creativity and academic performance (Alotaibi, 2025; Chiu & Hwang, 2025), but few empirical studies look into how GenAI affects moral psychological mechanisms and behavioral intentions. Also, there’s a lack of a rigorously validated standard tool in relevant communities to measure Moral Suspension in GenAI interactions. This hinders theoretical development and restricts GenAI evaluation and educational intervention implementation.

In this paper, our study focuses on Chinese emerging adults and aims to develop and validate the Moral Suspension Scale (MSS) for academic and creative use scenarios of GenAI. First, we clarify the core and structure of moral suspension through an interdisciplinary literature review, expert research, and surveys. Then, following standardized scale development procedures, we integrate theoretical insights with empirical findings to generate measurement items. After multiple rounds of data collection and analysis, we confirm the scale’s structural validity and reliability. Ultimately, we establish a scientific and reliable MSS for Chinese emerging adults.

This study expands the theoretical framework linking human-computer interaction (HCI) and moral psychology, extending traditional theories into the context of GenAI use. At a practical level, we developed a scientifically reliable Moral Suspension Scale. This scale serves not only as a tool for future evaluations of GenAI and its ethical impact but also provides developers, educators, and policymakers with a new perspective on identifying and addressing moral risks in technology use. It facilitates a deeper understanding of how Chinese emerging adults navigate moral judgment and responsibility attribution in specific scenarios such as AI-assisted writing, creative generation, and information processing. Furthermore, it offers valuable guidance for improving GenAI design and education strategies, advancing the goal of building human-centered GenAI systems.

## 2. Literature Review and related work

### 2.1. Novel ethical challenges in human-GenAI interaction

While the widespread adoption of GenAI has significantly improved efficiency and stimulated innovation, the complexity, instability, and opacity of the decision-making process have made the attribution of ethical responsibility more challenging than ever before (Bohacek & Farid, 2024; Tigard, 2021). Especially in higher education and the workplace scenario, many emerging adults are dependent on GenAI applications to complete tasks such as academic writing, research support, creative production, professional communication, and decision support. It has sparked ethical issues regarding plagiarism attribution and authorship responsibility conflicts (J. Luo, 2024; Waqas et al., 2025). Matthias (2004) first articulated the concept of the “*Responsibility Gap*,” noting that when autonomous machines cause immoral consequences, traditional accountability models face fundamental challenges, as manufacturers or operators often cannot be held accountable for consequences beyond their control or prediction.

Traditional theories of moral responsibility emphasize that free will and cognitive ability are fundamental prerequisites. For instance, the Aristotelian framework posits that the attribution of responsibility requires three conditions: voluntariness, cognitive awareness, and prudence (Constantinescu et al., 2022; Meyer, 2011). Many scholars argue that current AI systems lack free will/voluntariness (Sison & Redín, 2023) and the physiological basis necessary for perceiving suffering and understanding moral consequences (Bernáth, 2021; Himma, 2009). Consequently, their behavior is essentially a mechanical execution of predetermined programs, failing to fulfill the core criteria of responsibility. In contrast, Courtenage (2024) argues that systems with formal decision-making structures, value assessment mechanisms, and execution capabilities can assume “functional responsibility.” Floridi and Sanders (2004) propose a more disruptive “*Moral Agency without a Mind*” framework, asserting that three elements: interactivity, autonomy, and adaptability, are sufficient to establish the responsible status of artificial agents. This viewpoint has faced harsh criticism from Véliz (2021), who argues that while such entities can produce behavior compliant with ethical norms, they cannot understand the emotional consequences of their actions; it is essentially a technological machine with no ethical sense.

As traditional theories face controversy and challenges, the human-GenAI co-decision-making process also blurs the boundaries of responsibility. A significant reason for this phenomenon is that human-GenAI interaction generates the hybrid moral situations, where humans and AI function as co-agents in decision-making, thus forming an intertwined network of responsibility (Dong & Bocian, 2024). These hybrid situations diminish the intuitiveness of individual responsibility attribution and, combined with the opacity of AI decision-making, contribute to the emergence of a suspension phenomenon. In this context, users find themselves unable to fully justify the legitimacy of their actions while simultaneously grappling with the difficulty of completely attributing blame to the AI.

To address the responsibility ambiguity posed by these hybrid moral contexts, scholars have introduced the concepts of “*Answerability*” and “*Explainability*” (Arrieta et al., 2020; Tigard, 2021). Coeckelbergh (2020) argues that every decision output by AI systems should come with a traceable explanation process, enabling affected individuals to pose reasonable inquiries to the responsible parties. Arrieta et al. (2020) particularly emphasize that enhancing algorithmic transparency and model visualization can alleviate the barriers to responsibility attribution posed by the “black box” of AI. However, as Ross (2024) cautions, the fundamental objective of pursuing AI system transparency is to establish trust between stakeholders. Only by integrating mechanisms of transparency with social norms, value orientations, and multi-layered accountability frameworks can a just and equitable distribution of AI responsibility truly be realized.

Building on the preceding discussion, this study contends that the moral uncertainties inherent in human-GenAI interaction extend beyond mere difficulties in responsibility attribution. At a deeper level, these uncertainties manifest as a state of “*Suspension*” concerning users’ moral agency in uncertain and ambiguous contexts—where they find it challenging to either fully assume responsibility or completely absolve themselves from it (Malle, 2016; Waqas et al., 2025). To elucidate the psychological mechanisms and behavioral consequences associated with this passive state of “*Suspension*,” it is essential to develop new concepts and measurement tools that build upon the traditional frameworks. Such advancements would serve as vital guidance for the design of technologies and educational interventions.

## 2.2. Conceptualization of moral suspension

Currently, previous moral psychology frameworks show limitations in explaining the ethical ambiguity that arises in human-GenAI collaborative contexts. Existing theories of “*Moral Disengagement (excuses)*” and “*Diffusion of Responsibility*” emphasize how individuals employ various rationalization mechanisms, such as shifting responsibility or distorting outcomes, to evade moral pressures (Bandura, 1999a; Newman et al., 2020). These mechanisms collectively enable individuals to reinterpret inherently immoral actions as acceptable, shifting the perceived responsibility away from themselves or minimizing the harm caused (Dahl & Waltzer, 2018). For instance, by transferring or diffusing responsibility, individuals can attribute their actions to others, groups, or external systems, thus alleviating their psychological burden (Bleher & Braun, 2022).

However, decision-making scenarios involving GenAI are often characterized by high complexity, information asymmetry, and a lack of transparency in logic. In such contexts, individuals do not always actively seek to avoid responsibility; rather, they may find themselves in a state marked by dual ambiguity concerning moral attribution and behavioral norms (Malle et al., 2025). For example, experiments conducted by Jolly et al. (2026) indicate that individuals who engage with algorithms that restrict autonomy are more likely to commit unethical behaviors compared to those using autonomous algorithms. These individuals also struggle to entirely absolve themselves of responsibility toward authority figures (including AI, managers, or organizations). This condition, where individuals find it challenging to assume or reject responsibility, differs fundamentally from the typical moral disengagement mechanisms defined by Bandura (2017). Instead, it resembles a “grey area” of moral responsibility; thus, this study conceptualizes it as “*Moral Suspension*.” Table 1 illustrates the distinction between the concepts of “*Moral Suspension*” and “*Moral Disengagement*.”

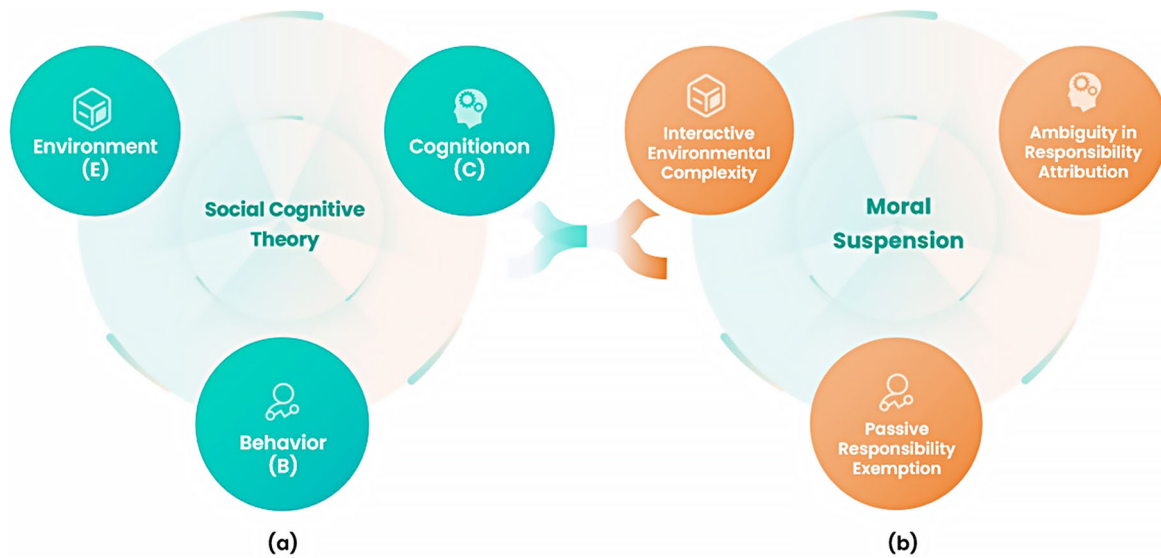
## 2.3. Theoretical framework of evaluating the moral Suspension: A social cognitive Theory perspective

Social Cognitive Theory, developed by psychologist Albert Bandura, emphasizes that individual behavior is not solely determined by internal cognition or external environments in a one-way manner. Instead, it is shaped by the continuous and dynamic interplay between environment, cognition, and behavior (Bandura, 1989, 1999b; Gao et al., 2009). From this perspective, the proposed concept of *Moral Suspension* can be further defined as a passive moral cognitive issue arising from the intertwining of three dimensions during human-GenAI interactions: 1) interactive environmental complexity, 2) ambiguity in responsibility attribution, and 3) passive responsibility exemption. The structural relationship among these three dimensions and their role in shaping moral suspension is shown in Figure 1.

In this framework, the complexity of the interaction environment corresponds to environmental factors, underscoring how the intricate nature of AI technology itself influences moral decision-making. The ambiguity of responsibility attribution aligns with individual factors, highlighting the psychological confusion and biases individuals experience in determining where responsibility lies. Lastly, the aspect of passive responsibility exemption relates to behavioral factors, emphasizing how individuals

**Table 1.** Conceptual discernment of moral Suspension and moral disengagement.

|                            | Moral Disengagement                                                                                                                                                                                                                                       | Moral Suspension                                                                                                                                                                                                                                                                                          |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Concept/ Definition</b> | Individuals actively use cognitive restructuring strategies (e.g., blame shifting, consequence downplaying, moral justification, etc.) to alleviate guilt and psychological stress resulting from unethical behaviours and to achieve self-justification. | The middle ground of moral attribution in human-GenAI interactions, caused by the intersection of technological complexity, ambiguity of attribution of responsibility, and passive exoneration of responsibility, which makes it “difficult to take responsibility and difficult to get out of it.”      |
| <b>Core mechanism</b>      | Actively avoiding moral pressure through rationalization mechanisms, perceiving immoral behavior as acceptable, causing negligible harm, or not one’s responsibility, emphasizing the subject’s proactivity and self-defense.                             | The complexity of the environment limits access to information, responsibility is unclear, and individual responsibility is passively weakened. This is not a case of actively shirking responsibility, but rather a blurring of moral responsibility caused by the combined effects of multiple factors. |
| <b>Key literature</b>      | (Bandura, 1989, 1999a, 2017; Dahl & Waltzer, 2018)                                                                                                                                                                                                        | (S. J. Bennett, 2025; Malle, 2016; Malle et al., 2025; Waqas et al., 2025)                                                                                                                                                                                                                                |



**Figure 1.** a) Social cognitive Theory; b) moral Suspension framework.

demonstrate a tendency to relinquish responsibility when AI is involved. The theoretical contribution of moral suspension lies in its ability to transcend the binary opposition between ontology (which posits that consciousness is a prerequisite for responsibility) and functionalism (which argues that social relational practices can establish responsibility) (Coeckelbergh, 2020; Véliz, 2021). This framework provides a fresh lens through which to understand and analyze the ethical uncertainties inherent in human-GenAI collaboration, opening avenues for further exploration into the complexities of moral decision-making in technologically mediated environments.

### 2.3.1. Interactive environmental complexity

This dimension arises from the inherent opacity and autonomy of GenAI, which impose significant cognitive and behavioral pressures on individuals (Hassija et al., 2024; Jo & Raj, 2024; Marcus & Teuwen, 2024; Pafla et al., 2024). Such opacity severely undermines users' ability to assess the potential moral risks associated with their actions (Brožek et al., 2024; Ross, 2024). From the perspective of technological moral virtues, when users cannot comprehend the logic of AI systems, their moral capacity to judge the propriety of those actions is obstructed, often leading to a temporary suspension or standstill in moral judgment (Khalili, 2024). Moreover, the increasing agency of GenAI exacerbates the complexity of the environment (Bigman & Gray, 2018; Coeckelbergh, 2023), further diminishing human control. This phenomenon is particularly pronounced in areas such as creative generation, academic writing, and intelligent diagnostics (Hinostroza Fuentes et al., 2025; Hull & Fins, 2024). In summary, the complexity of the interaction environment constitutes a foundational external pressure of moral suspension. By limiting accessibility to information and the transparency of processes, it directly interferes with users' ability to form clear moral judgments (Arrieta et al., 2020; H. Xu & Shuttleworth, 2024).

### 2.3.2. Ambiguity in responsibility attribution

This dimension focuses on the cognitive ambiguity arising from the difficulty of clearly defining the agency and contribution of different parties in human-GenAI collaboration. In situations where GenAI participates in decision-making, the attribution of responsibility often becomes ambiguous. According to moral psychology theories, the assignment of responsibility for actions depends on users' perceptions of the agency and mental states of the actors involved (Gray et al., 2012; Shank et al., 2019). Gray and Wegner (2009) argue that only those perceived as moral agents with a will can be held accountable for their actions, while entities lacking cognitive abilities are typically regarded as "moral patients" who bear no responsibility. However, the involvement of AI as an actor complicates this attribution process,

which can lead to biases and uncertainties (Bleher & Braun, 2022; Yam et al., 2022). The moral roles in such interactions become unclear: Is AI an autonomous agent, or is it merely functioning as a tool? (Sundar, 2020; Swanepoel, 2021). If AI is not considered a genuine moral agent, then traditional frameworks of responsibility would primarily assign accountability to humans (Awad et al., 2020). Conversely, if individuals attribute some degree of “mind” to AI, they may be inclined to share responsibility with the AI (Joo, 2024; Ladak et al., 2024; Waytz et al., 2010). This leads to ambiguity and disagreement in responsibility attribution, leaving room for varied moral interpretations (Dana et al., 2007). The implications of this ambiguity are significant, as they can affect not only how individuals perceive their ethical obligations in human-GenAI interactions but also how accountability is managed in contexts such as healthcare, autonomous systems, and automated decision-making processes. Ultimately, the blurred lines of responsibility in these scenarios contribute to the phenomenon of moral suspension, as individuals grapple with the shifting dynamics of accountability and the extent to which AI should be considered a moral actor.

### **2.3.3. Passive responsibility exemption**

This dimension reflects a passive erosion of individuals’ perceptions of their own responsibility, leading to an unintended diminished sense of accountability (Chen et al., 2025; Malle, 2016). Rather than actively rejecting responsibility, users exhibit a diminished sense of obligation through inaction, reliance on system autonomy, and attribution of blame to the technology itself (S. J. Bennett, 2025; Smit et al., 2025). Users may adopt a technological determinism perspective, attributing negative outcomes primarily or entirely to the limitations of GenAI, rather than acknowledging their own negligence, misjudgments, or improper instructions (Nyholm & Smids, 2016; Stahl, 2006). Hallström (2022) emphasizes the necessity of reassessing technological determinism in education and practice, as an overemphasis on the power of technology can undermine personal responsibility awareness. Additionally, mechanisms of asymmetric power and responsibility contribute to the passive exemption from accountability. In many human-GenAI collaborative contexts, individuals may have ultimate authority or decision-making power over AI-driven outcomes; however, when results are unfavorable, they often lack corresponding mechanisms to accept responsibility (Brailsford et al., 2025; Passlack et al., 2024; Schoenherr & Thomson, 2024). Research by Dong and Bocian (2024) further indicates that in moral decision-making situations involving both humans and AI, individuals tend to evaluate their unethical behavior more leniently if they can attribute part of the responsibility to the AI advisor and if their own misconduct aligns with the AI’s recommendations. This passive responsibility exemption can lead to a series of ethical challenges, including a diminished sense of moral agency and accountability. Individuals may find it easier to absolve themselves of wrongdoing when they can shift part of the blame to the technology, thus eroding the ethical frameworks that are necessary for responsible decision-making in human-GenAI interactions. As such, addressing this phenomenon is critical to fostering a culture of ethical responsibility in the age of advanced AI systems.

In summary, moral suspension is not merely a straightforward extension or terminological substitution of the theory of moral disengagement; there are essential differences between the two concerning their mechanisms, psychological processes, and behavioral outcomes (Bonner et al., 2016). More importantly, the three dimensions form the core experience of Moral Suspension. Therefore, it represents a distinctive passive moral cognitive issue deeply embedded in specific technological contexts, posing unprecedented challenges to traditional ethical reasoning models that are grounded in notions of human autonomy and information transparency (Malle, 2016). Understanding and evaluating moral suspension phenomena is crucial to better analyze the ethical implications of human-GenAI interaction and to promote the design of GenAI.

## **3. Development and validation process of the moral Suspension scale (MSS)**

Our research process strictly adhered to the standardized procedures for constructing psychometric instruments (Carpenter, 2018; Polit & Beck, 2006), employing a progressive five-stage validation framework for the development and validation of the scale, including three rounds of independent data collection (Greco et al., 2021; Keng et al., 2025).

In Phase I, we focused on establishing the theoretical grounding of relevant concepts and generating initial items, ensuring a robust theoretical foundation for the scale design. In Phase II, we conducted a content validity assessment and a small-scale pilot test to screen and optimize the items. Phase III involved item analysis and exploratory factor analysis (EFA) to examine the internal structure of the scale. Phase IV comprised confirmatory factor analysis (CFA) and criterion validity testing to confirm the scale's factor structure and external validity. Finally, measurement invariance was tested, and the finalized scale was established in Phase V.

### 3.1. Applicable scenarios

This scale is intended to measure moral suspension in Chinese emerging adults during AI-assisted academic/work-related writing, research support, creative production, professional communication, and decision support tasks. As digital natives at the intersection of higher education and early career development, this group frequently uses GenAI tools (such as ChatGPT, Deepseek, etc.) for content generation, literature processing, and language optimization. These tasks often involve ethically ambiguous situations, including covert plagiarism, factual distortion, biased information, and data fabrication, which pose unique challenges to moral judgment and responsibility attribution. The scale is thus calibrated to assess psychological states of moral uncertainty in such scenarios, where GenAI participation is functionally embedded in users' academic or professional workflows.

#### 3.1.1. Target population

The target population for our study was defined by specific screening criteria: (1) participants were required to be between 18 and 29 years old; and (2) they must have prior experience using GenAI tools (such as ChatGPT, Deepseek, etc.) to complete academic writing or work-related tasks. At the beginning of the questionnaire, participants were presented with a scenario involving AI-assisted writing and were asked, based on their own experience and subjective perceptions, to make judgments and respond to morally ambiguous situations arising from the use of GenAI tools in processes such as content generation, literature processing, and language optimization, including issues like covert plagiarism, factual distortion, biased output, or fabricated data.

#### 3.1.2. Data collection

We employed a multi-phase random sampling strategy, with three rounds of data gathering (Phase II:  $n = 76$ ; Phase III:  $n = 346$ ; Phase IV:  $n = 363$ ) conducted between July and November 2025 via the professional online survey platform *HuiXiang Data*. All participants were native Chinese speakers, and the survey items were administered in Simplified Chinese. Each round excluded individuals who had already participated, ensuring that each participant completed the questionnaire only once across all three rounds. In addition to this, strict data exclusion criteria were applied to promote data quality. Responses were excluded if participants: (1) failed attention check questions (trap items); (2) completed the survey in an unreasonably short time, defined as a total duration of less than 200 s (in the pretest, the simulated response time consistently exceeded 200 s), which indicates insufficient cognitive processing; (3) failed the tool verification check, where participants provided irrelevant answers or non-GenAI tools (e.g., search engines) in the mandatory open-ended question regarding tool usage; (4) exhibited logical inconsistency, specifically those claiming more than 3 years of GenAI experience, which conflicts with the release timeline of mainstream generative AI models (e.g., ChatGPT). The platform distributed the survey link to a sample pool of 1.5 million registered users to recruit eligible respondents. The study has been approved by the Institutional Review Board of The Hong Kong Polytechnic University. All participants read an informed consent statement and agreed to participate voluntarily before completing the survey. Each participant received a reward of 4 RMB upon successful completion of the questionnaire.

### 3.2. Phase I: Construct definition and item generation

To ensure a solid theoretical foundation and high content validity for the scale, this study drew on the triadic reciprocal model of social cognitive theory (Bandura, 1999b) as well as recent research in AI ethics (Please see Sections 2.2 & 2.3). In this context, the concept of Moral Suspension was conceptualized as a passive state of moral ambiguity comprising three core dimensions: Interactional Environmental Complexity, Ambiguity in Responsibility Attribution, and Passive Responsibility Exemption. This construct is theoretically distinguished from the more active cognitive strategy of moral disengagement (A. Luo & Bussey, 2023; Newman et al., 2020).

Building on this clear conceptual boundary, item development was undertaken using a deductive strategy that integrated theoretical deduction (deriving constructs from interdisciplinary literature) with scenario-based operationalization (deducing relevant contexts from empirical findings). Regarding the theoretical deduction, items for the *Interactive Environmental Complexity* dimension were adapted from research on the black box problem in technology (Adadi & Berrada, 2018; Brožek et al., 2024; Hassija et al., 2024; Minh et al., 2022); items for the *Ambiguity in Responsibility Attribution* dimension drew on attribution theory in HCI (Bleher & Braun, 2022; Horstmann & Krämer, 2022; Moon & Nass, 1998; Pavone et al., 2023); and items for the *Passive Responsibility Exemption* dimension were grounded in the mechanism of passive responsibility abdication attributed to technological determinism (S. J. Bennett, 2025; Campolo & Crawford, 2020; Hallström, 2022; Johnson, 2015; Rowe et al., 2024; Ryoo et al., 2024). Regarding the scenario-based operationalization, grounded in empirical findings on GenAI usage patterns, we identified specific high-stakes scenarios, such as handling factual fabrications in academic writing (e.g., Item ENV7: “AI exhibits randomness and uncertainty when generating content”) (J. Luo, 2024), navigating ethical ambiguities in creative generation (e.g., Item BEH9: “When AI output is ethically unclear, I usually accept it without questioning”) (S. J. Bennett, 2025; Weber et al., 2024), and determining accountability in collaborative decision-making (e.g., Item COG12: “When facing ambiguous AI outputs, I find it difficult to distinguish the boundary of responsibility between myself and AI”) (Passlack et al., 2024; Schoenherr & Thomson, 2024). Based on both theoretical constructs and the real-world GenAI usage scenarios of Chinese emerging adults, we developed preliminary measurement items for each dimension, which constituted the initial item pool. Ultimately, in this first stage, the study generated a set of 45 initial items, covering scenarios related to academic and work-related pressures (such as AI-enabled covert plagiarism or biased outputs during manuscript preparation). To mitigate acquiescence bias and ensure response validity, 3 reverse-coded items were incorporated into each of the 3 dimensions. Consequently, the final pool included 9 reverse-coded items, constituting 20% of the total item pool, which aligns with methodological recommendations for balancing scale polarity (DeVellis & Thorpe, 2021; Weijters & Baumgartner, 2012). Table 2 presents the dimensions and item information of the MSS.

### 3.3. Phase II: Content validity and pilot testing

To ensure adequate content validity of the scale, a panel of 12 experts was convened, all with backgrounds in HCI ( $n=5$ ), moral psychology ( $n=3$ ), and psychometric statistics ( $n=4$ ). These experts were asked to evaluate the initial pool of 45 items using a 4-point scale (1 = not relevant, 4 = highly relevant) to assess the Content Validity Index (CVI), which reflects the degree to which each item matches its defined dimension. Subsequently, a critical value screening approach ( $I\text{-CVI} \geq 0.78$ ,  $S\text{-CVI} \geq 0.90$ ) was employed to filter out non-essential items (Polit & Beck, 2006). Results showed that all items had an  $I\text{-CVI}$  above 0.9, and the average  $S\text{-CVI}$  reached 0.987, both surpassing the recommended thresholds of 0.78 and 0.9. This indicates that the scale demonstrated a high level of content representativeness and dimensional relevance among the expert panel.

Based on the expert feedback, we thoroughly discussed the phrasing of each item and refined the wording or content where appropriate, while retaining the original meaning. For example, for the first dimension (Interactive Environmental Complexity) and the second dimension (Ambiguity in Responsibility Attribution), all item subjects were standardized and foregrounded (using “AI” and “I” respectively as the subjects) to reduce ambiguity between these two dimensions. For the third

**Table 2.** Dimensions and item distribution of the MSS.

| Dimension                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                 | References                                                                                                            | Item number                           |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Interactive Environmental Complexity    | Items under this dimension primarily address the complexity, opacity, and difficulty of understanding the decision-making processes of AI systems. The focus is on how the technological environment influences users' moral judgment and responsibility attribution. For example, <i>"AI lacks transparency in the process of result generation."</i>                                                                                      | (Adadi & Berrada, 2018; Brožek et al., 2024; Hassija et al., 2024; Minh et al., 2022)                                 | 15 (including 3 reverse-scored items) |
| Ambiguity in Responsibility Attribution | Items under this dimension focus on users' difficulties in clearly identifying the responsible party and the boundaries of responsibility in contexts involving AI participation. The emphasis is on the uncertainty experienced by the cognitive subject regarding the division of responsibility. For example, <i>"The longer I use AI tools, the less clear my recognition of my own moral responsibility becomes."</i>                  | (Bleher & Braun, 2022; Horstmann & Krämer, 2022; Moon & Nass, 1998; Pavone et al., 2023)                              | 15 (including 3 reverse-scored items) |
| Passive Responsibility Exemption        | Items under this dimension are designed to capture the passive weakening of users' sense of responsibility and the ambiguity of responsibility awareness caused by the automaticity and influence of AI. The emphasis is on the passive reduction of responsibility cognition after the occurrence of behavioral outcomes. For example, <i>"When AI output seems questionable, action can wait because the system will correct itself."</i> | (S. J. Bennett, 2025; Campolo & Crawford, 2020; Hallström, 2022; Johnson, 2015; Rowe et al., 2024; Ryoo et al., 2024) | 15 (including 3 reverse-scored items) |

dimension (Passive Responsibility Exemption), all item subjects were crafted to emphasize passivity, thereby aligning more closely with the core meaning of the dimension itself.

After confirming content validity, we conducted the first small-scale pilot test to assess the comprehensibility and practicality of the scale in real applications. A total of 100 questionnaires were collected, resulting in 76 valid responses (validity rate: 76%) from Chinese emerging adults with varying socioeconomic backgrounds (e.g., education, major, ethnicity, hometown, and family income) and AI usage experiences (see Table 3). Through statistical analysis of participants' response distributions, missing data rates, and response times, we further identified items that presented comprehension difficulties or ambiguities, which were then revised accordingly. Additionally, trap items were included for quality control. As a result, by the end of the second stage, the scale comprised 40 items (including 8 reverse-scored items; excluding a trap item), with 14 items in the Interactive Environmental Complexity dimension, 12 in the Ambiguity in Responsibility Attribution dimension, and 14 in the Passive Responsibility Exemption dimension.

### 3.4. Phase III: Item analysis and exploratory factor analysis (EFA)

To examine the psychometric properties of the Moral Suspension Scale developed in Phase II, the third round of data collection yielded 346 valid responses from 400 participants (validity rate: 86.5%), which were used for this phase (see Table 4). The internal consistency reliability was evaluated using both Cronbach's  $\alpha$  (Holmbeck & Devine, 2009) and the corrected item-total correlation (CITC) (Cristobal et al., 2007). Data analysis showed that the Cronbach's  $\alpha$  coefficients for the three dimensions were 0.914, 0.909, and 0.938, respectively—all exceeding the commonly accepted threshold of 0.70. Based on the CITC cutoff value (0.4), one item in the Ambiguity in Responsibility Attribution (COG6, CITC = 0.326) did not meet the standard and was thus removed, resulting in 39 items being retained for the EFA.

As principal component analysis is not appropriate for scale development, our study employed the principal axis factoring method for EFA (Carpenter, 2018). The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy (KMO = 0.939) and Bartlett's Test of Sphericity ( $p < 0.001$ ) confirmed the

**Table 3.** Participant characteristics in the pilot test ( $n = 76$ ).

| Characteristic      | Items                          | Frequency | Percentage (%) |
|---------------------|--------------------------------|-----------|----------------|
| Gender              | Male                           | 30        | 39.5           |
|                     | Female                         | 46        | 60.5           |
| Age ( $M \pm SD$ )  | 24.68 $\pm$ 2.68               |           |                |
| Education           | High school/vocational school  | 5         | 6.6            |
|                     | Undergraduate/Associate degree | 57        | 75.0           |
|                     | Master's degree                | 11        | 14.5           |
|                     | Doctoral degree                | 3         | 3.9            |
| Ethnicity           | Han Chinese                    | 67        | 88.2           |
|                     | Ethnic Minorities              | 9         | 11.8           |
| Major               | STEM & Medical Sciences        | 35        | 46.1           |
|                     | Social Sciences                | 28        | 36.8           |
|                     | Humanities & Arts              | 13        | 17.1           |
| Hometown            | First-tier Cities              | 16        | 21.0           |
|                     | Second-tier Cities             | 27        | 35.5           |
|                     | Third-tier Cities              | 10        | 13.2           |
|                     | County Cities                  | 13        | 17.1           |
|                     | Rural Areas                    | 10        | 13.2           |
| Family Income       | < 50,000                       | 3         | 3.9            |
|                     | 50,000–100,000                 | 12        | 15.9           |
|                     | 100,000–200,000                | 40        | 52.6           |
|                     | 200,000–500,000                | 19        | 25.0           |
|                     | > 500,000                      | 2         | 2.6            |
| AI Usage Experience | Less than 3 months             | 3         | 3.9            |
|                     | 3–6 months                     | 9         | 11.8           |
|                     | 6–12 months                    | 24        | 31.5           |
|                     | 1–2 years                      | 20        | 26.3           |
|                     | 2–3 years                      | 20        | 26.3           |

suitability of the data for factor analysis (Pett et al., 2003). The criteria for factor extraction (Keng et al., 2025) included: eigenvalues greater than 1.0 (the Kaiser-Guttman rule), factor loadings greater than 0.50, and communalities greater than 0.40 (Costello & Osborne, 2005). Items failing to meet these thresholds were considered for exclusion to maintain sufficient variance contribution. After seven iterations of varimax rotation, three factors with eigenvalues above 1.0 were extracted, cumulatively explaining 49.81% of the variance, aligning with the theoretically hypothesized three-dimensional structure. The scree plot test also supported the extraction of three factors. One item (ENV8) was excluded due to a factor loading below the cutoff value of 0.50, resulting in a final scale of 38 items for the Moral Suspension Scale. Table 5 presents the communalities and the full factor loading matrix for all 38 items. The results indicated that all retained items exhibited strong primary loadings on their intended constructs with no problematic cross-loadings.

### 3.5. Phase IV: Scale validation and confirmatory factor analysis (CFA)

To further examine the structural validity identified in the EFA, a total of 363 valid samples from 400 participants (validity rate: 90.8%) in the third round of data collection were subjected to confirmatory factor analysis (CFA) using AMOS 24.0 (see Table 6). The three-factor model, consisting of 38 items derived from EFA, was tested. In the CFA, all items were found to have factor loadings greater than 0.5 (Bagozzi & Yi, 1988), confirming that each item was a satisfactory indicator of its intended latent construct and thus all were retained.

Next, a comprehensive evaluation of model fit was conducted using several widely adopted indices: the chi-square to degrees of freedom ratio ( $\chi^2/df$ ), Goodness-of-Fit Index (GFI), Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Incremental Fit Index (IFI), Root Mean Square Error of Approximation (RMSEA), Normed Fit Index (NFI), and Standardized Root Mean Square Residual (SRMR). According to established conventions, a well-fitting model is typically indicated by a  $\chi^2/df$

**Table 4.** Participant characteristics in the Phase III ( $n = 346$ ).

| Characteristic      | Items                          | Frequency | Percentage (%) |
|---------------------|--------------------------------|-----------|----------------|
| Gender              | Male                           | 133       | 38.4           |
|                     | Female                         | 213       | 61.6           |
| Age ( $M \pm SD$ )  | 24.61 $\pm$ 2.74               |           |                |
| Education           | High school/vocational school  | 7         | 2.0            |
|                     | Undergraduate/Associate degree | 283       | 81.8           |
|                     | Master's degree                | 53        | 15.3           |
|                     | Doctoral degree                | 3         | 0.9            |
| Ethnicity           | Han Chinese                    | 318       | 91.9           |
|                     | Ethnic Minorities              | 28        | 8.1            |
| Major               | STEM & Medical Sciences        | 165       | 47.7           |
|                     | Social Sciences                | 117       | 33.8           |
|                     | Humanities & Arts              | 64        | 18.5           |
| Hometown            | First-tier Cities              | 74        | 21.4           |
|                     | Second-tier Cities             | 107       | 30.9           |
|                     | Third-tier Cities              | 83        | 24.0           |
|                     | County Cities                  | 42        | 12.1           |
|                     | Rural Areas                    | 40        | 11.6           |
| Family Income       | < 50,000                       | 15        | 4.3            |
|                     | 50,000–100,000                 | 69        | 20.0           |
|                     | 100,000–200,000                | 161       | 46.5           |
|                     | 200,000–500,000                | 94        | 27.2           |
|                     | > 500,000                      | 7         | 2.0            |
| AI Usage Experience | Less than 3 months             | 8         | 2.3            |
|                     | 3–6 months                     | 43        | 12.4           |
|                     | 6–12 months                    | 93        | 26.9           |
|                     | 1–2 years                      | 143       | 41.3           |
|                     | 2–3 years                      | 59        | 17.1           |

Note: Using the G\*Power 3.1.9.2 software, statistical power analysis was conducted (with  $\alpha = 0.05$ , power = 0.95, effect size  $f^2 = 0.15$ ), determining that the minimum required sample size was 290 for the survey in Phase III. The actual sample size exceeded the thresholds ( $n = 346 > 290$ ), thereby ensuring the statistical validity of the data analyses and the robustness of the study's conclusions.

value less than 3, GFI, CFI, TLI, IFI, and NFI values greater than 0.90, and RMSEA and SRMR values less than 0.08 (Bagozzi & Yi, 1988; Chen et al., 2023; Keng et al., 2025).

As shown in Table 7, the results of the CFA revealed a  $\chi^2$  value of 1196.124 with 662 degrees of freedom ( $p < 0.001$ ), yielding a  $\chi^2/\text{df}$  ratio of 2.078, which is below the threshold of 3. Furthermore, the CFI (0.949), TLI (0.946), and IFI (0.949) all exceeded the ideal cutoff of 0.90. The RMSEA (0.047) and SRMR (0.041) were both substantially below the 0.08 criterion. Although the GFI (0.855) and NFI (0.893) were slightly below the recommended value of 0.90, all other core indices demonstrated excellent fit. Collectively, these results indicate that the three-factor model fits the observed data well and that the structural validity of the scale is well supported.

In terms of convergent validity, our study assessed both Composite Reliability (CR) and Average Variance Extracted (AVE). As shown in Table 8, the results indicated that the CR values for the three factors were 0.936, 0.954, and 0.960, respectively, all of which are substantially higher than the recommended threshold of 0.70, demonstrating high internal consistency for the scale. For AVE, the value for Interactive Environmental Complexity (Factor 1), Ambiguity in Responsibility Attribution (Factor 2), and Passive Responsibility Exemption (Factor 3) were 0.529, 0.654 and 0.635, exceeding the ideal criterion of 0.5 (Huang et al., 2013; Keng et al., 2025). Therefore, the convergent validity of the scale is confirmed.

Regarding discriminant validity, our study followed the criterion proposed by Fornell and Larcker (1981), which involves comparing the square root of the AVE for each factor with the correlation coefficients between that factor and the others. As shown in Table 9, the results showed that the square roots of the AVE for the three factors (0.727, 0.809, 0.797; on the diagonal) were all greater than the corresponding inter-factor correlations (off-diagonal values). This indicates that each dimension demonstrated good discriminant validity, with each construct capturing a distinct concept. Furthermore, the

**Table 5.** Rotated factor loadings for the 38-item MSS ( $n = 346$ ).

| Dimension after EFA                          | Original code | Item                                                                                                                       | Factor 1 | Factor 2 | Factor 3 | Communality |
|----------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-------------|
| Interactive Environmental Complexity (13)    | ENV12         | AI fails to clarify the standards for tracing and verifying its generated content.                                         | 0.709    | 0.301    | −0.014   | 0.638       |
|                                              | ENV3          | AI lacks transparency in the process of result generation.                                                                 | 0.696    | 0.196    | 0.022    | 0.564       |
|                                              | ENV4          | AI prevents the specific sources of potential problems in its outputs from being identified.                               | 0.670    | 0.277    | −0.007   | 0.547       |
|                                              | ENV14         | AI automatically flags risk points in its output content.                                                                  | 0.658    | 0.240    | 0.017    | 0.541       |
|                                              | Reverse ENV1  | AI makes it difficult to understand the underlying logic and reasons behind its generated content.                         | 0.648    | 0.101    | 0.009    | 0.478       |
|                                              | ENV2          | AI produces noticeably different outputs when executing the same instructions repeatedly.                                  | 0.638    | 0.113    | 0.091    | 0.494       |
|                                              | ENV6          | AI provides a detailed explanation of its content generation process on the user interface.                                | 0.634    | 0.134    | 0.039    | 0.462       |
|                                              | Reverse ENV11 | AI keeps the underlying data and logic of its generated content invisible.                                                 | 0.628    | 0.266    | 0.001    | 0.532       |
|                                              | ENV7          | AI exhibits randomness and uncertainty when generating content.                                                            | 0.624    | 0.163    | 0.087    | 0.499       |
|                                              | ENV9          | AI makes updates and changes within the system invisible to users.                                                         | 0.598    | 0.262    | 0.036    | 0.484       |
|                                              | ENV5          | AI conceals potential moral risks during interactions.                                                                     | 0.587    | 0.343    | −0.015   | 0.524       |
|                                              | ENV13         | AI lacks quality assurance measures when confronted with unfamiliar problems.                                              | 0.582    | 0.336    | −0.049   | 0.524       |
|                                              | ENV10         | AI provides users with aids to help understand the interaction process.                                                    | 0.581    | 0.196    | 0.025    | 0.441       |
| Ambiguity in Responsibility Attribution (11) | Reverse COG12 | When facing ambiguous AI outputs, I find it difficult to distinguish the boundary of responsibility between myself and AI. | 0.306    | 0.721    | 0.159    | 0.635       |
|                                              | COG2          | If I fail to detect moral issues in AI content in time, I am uncertain whether I should be held responsible afterwards.    | 0.335    | 0.678    | 0.087    | 0.615       |
|                                              | COG11         | The longer I use AI tools, the less clear my recognition of my own moral responsibility becomes.                           | 0.233    | 0.678    | 0.125    | 0.633       |
|                                              | COG5          | When faced with unethical conclusions from AI, I find it difficult to generate a clear sense of responsibility.            | 0.290    | 0.660    | 0.107    | 0.569       |
|                                              | COG8          | When using AI, I feel my sense of moral responsibility gradually becomes blurred.                                          | 0.220    | 0.660    | 0.191    | 0.611       |
|                                              | COG9          | I usually cannot judge whether the ambiguous information generated by AI deserves to be handled responsibly by me.         | 0.176    | 0.657    | 0.186    | 0.537       |
|                                              | COG3          | For hidden moral issues in AI content (such as implicit bias), I cannot clearly determine who is at fault.                 | 0.331    | 0.636    | 0.115    | 0.581       |
|                                              | Reverse COG4  | When facing the results of moral risks generated in AI interaction, I feel that responsibility can be clearly defined.     | 0.363    | 0.614    | 0.118    | 0.556       |
|                                              | COG10         | When facing controversial content generated by AI, I doubt the extent to which I should bear moral responsibility.         | 0.276    | 0.608    | 0.080    | 0.491       |
|                                              | COG7          | When AI generates ambiguous information, I feel the attribution of responsibility is clear.                                | 0.332    | 0.587    | 0.061    | 0.512       |
|                                              | Reverse COG1  | When using AI, I find it difficult to clearly judge how much moral responsibility I should bear.                           | 0.349    | 0.582    | 0.111    | 0.519       |
| Passive Responsibility Exemption (14)        | BEH8          | I don't typically reflect deeply on the potential ethical risks in AI content.                                             | 0.018    | 0.116    | 0.824    | 0.729       |
|                                              | BEH10         | I tend to trust that the AI is designed ethically, so I don't feel the need to monitor it.                                 | −0.024   | 0.044    | 0.814    | 0.671       |
|                                              | BEH9          | When AI output is ethically unclear, I usually accept it without questioning.                                              | −0.028   | 0.157    | 0.797    | 0.674       |
|                                              | BEH7          | I rarely think about what ethical principles the AI follows when I use it.                                                 | 0.062    | 0.066    | 0.791    | 0.684       |
|                                              | BEH12         | I tend to trust the AI's default settings to handle ethical issues, so I skip additional checks.                           | 0.040    | 0.098    | 0.789    | 0.700       |
|                                              | BEH11         | I see it as beyond my duty to give feedback or intervene when I spot ethical risks in AI content.                          | 0.038    | 0.150    | 0.782    | 0.670       |
|                                              | BEH13         | In uncertain situations, I usually don't take action to prevent ethical problems with AI.                                  | −0.016   | 0.047    | 0.746    | 0.636       |

(continued)

**Table 5.** Continued.

| Dimension after EFA  | Original code | Item                                                                                                                            | Factor 1 | Factor 2 | Factor 3 | Communality |
|----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-------------|
|                      | BEH14         | I consider the potential ethical risks posed by AI to be serious.                                                               | −0.022   | 0.054    | 0.731    | 0.592       |
|                      | Reverse       |                                                                                                                                 |          |          |          |             |
|                      | BEH2          | I don't usually make an effort to verify the ethical risks of AI-generated information.                                         | 0.009    | 0.172    | 0.719    | 0.612       |
|                      | BEH3          | When AI content has ethical controversies, I don't feel the need to adjust or correct it.                                       | 0.053    | 0.044    | 0.676    | 0.536       |
|                      | BEH6          | I trust the platform to handle any serious ethical issues, so I don't need to be concerned.                                     | 0.049    | −0.053   | 0.657    | 0.543       |
|                      | BEH1          | When AI output seems questionable, action can wait because the system will correct itself.                                      | 0.076    | 0.037    | 0.616    | 0.505       |
|                      | BEH4          | If errors arise from AI suggestions, I consider myself responsible for the outcome rather than the tool.                        | 0.100    | 0.180    | 0.532    | 0.523       |
|                      | Reverse       |                                                                                                                                 |          |          |          |             |
|                      | BEH5          | I believe that users bear the responsibility for addressing AI's ethical problems, rather than leaving it solely to developers. | 0.130    | 0.105    | 0.508    | 0.565       |
|                      | Reverse       |                                                                                                                                 |          |          |          |             |
| Eigen value          |               |                                                                                                                                 | 11.501   | 7.063    | 2.318    |             |
| % Variance explained |               |                                                                                                                                 | 29.491   | 18.11    | 5.943    |             |
| Cronbach's $\alpha$  |               |                                                                                                                                 | 0.914    | 0.909    | 0.938    |             |

**Table 6.** Participant characteristics in the Phase IV ( $n = 363$ ).

| Characteristic      | Items                          | Frequency | Percentage (%) |
|---------------------|--------------------------------|-----------|----------------|
| Gender              |                                |           |                |
|                     | Male                           | 149       | 41.0           |
|                     | Female                         | 214       | 59.0           |
| Age ( $M \pm SD$ )  | 24.51 $\pm$ 2.82               |           |                |
| Education           |                                |           |                |
|                     | High school/vocational school  | 4         | 1.1            |
|                     | Undergraduate/Associate degree | 295       | 81.3           |
|                     | Master's degree                | 61        | 16.8           |
|                     | Doctoral degree                | 3         | 0.8            |
| Ethnicity           |                                |           |                |
|                     | Han Chinese                    | 324       | 89.3           |
|                     | Ethnic Minorities              | 39        | 10.7           |
| Major               |                                |           |                |
|                     | STEM & Medical Sciences        | 173       | 47.7           |
|                     | Social Sciences                | 126       | 34.7           |
|                     | Humanities & Arts              | 64        | 17.6           |
| Hometown            |                                |           |                |
|                     | First-tier Cities              | 72        | 19.8           |
|                     | Second-tier Cities             | 112       | 30.9           |
|                     | Third-tier Cities              | 84        | 23.1           |
|                     | County Cities                  | 53        | 14.6           |
|                     | Rural Areas                    | 42        | 11.6           |
| Family Income       |                                |           |                |
|                     | < 50,000                       | 22        | 6.0            |
|                     | 50,000–100,000                 | 70        | 19.3           |
|                     | 100,000–200,000                | 188       | 51.8           |
|                     | 200,000–500,000                | 77        | 21.2           |
|                     | > 500,000                      | 6         | 1.7            |
| AI Usage Experience |                                |           |                |
|                     | Less than 3 months             | 10        | 2.8            |
|                     | 3–6 months                     | 43        | 11.8           |
|                     | 6–12 months                    | 115       | 31.7           |
|                     | 1–2 years                      | 128       | 35.2           |
|                     | 2–3 years                      | 67        | 18.5           |

Note: Using the G\*Power 3.1.9.2 software, statistical power analysis was conducted (with  $\alpha = 0.05$ , power = 0.95, effect size  $f^2 = 0.15$ ), determining that the minimum required sample size was 287 for the survey in Phase IV. The actual sample size exceeded the thresholds ( $n = 363 > 287$ ), thereby ensuring the statistical validity of the data analyses and the robustness of the study's conclusions.

**Table 7.** The model fit of the MSS.

| Index           | $\chi^2/df$ | RMSEA | SRMR  | GFI   | CFI   | TLI   | IFI   | NFI   |
|-----------------|-------------|-------|-------|-------|-------|-------|-------|-------|
| Value           | 1.807       | 0.047 | 0.041 | 0.855 | 0.949 | 0.946 | 0.949 | 0.893 |
| Reference value | <3.00       | <0.08 | <0.08 | >0.90 | >0.90 | >0.90 | >0.90 | >0.90 |

**Table 8.** CFA of the MSS ( $n = 363$ ).

| Dimension after EFA                                                                             | Original code | Item                                                                                                                           | CFA loading |
|-------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------|-------------|
| Interactive Environmental Complexity (Cronbach's $\alpha = 0.935$ , CR = 0.936, AVE = 0.529)    | ENV1          | AI makes it difficult to understand the underlying logic and reasons behind its generated content                              | 0.693       |
|                                                                                                 | ENV11         | AI keeps the underlying data and logic of its generated content invisible                                                      | 0.743       |
|                                                                                                 | ENV12         | AI fails to clarify the standards for tracing and verifying its generated content                                              | 0.759       |
|                                                                                                 | ENV13         | AI lacks quality assurance measures when confronted with unfamiliar problems                                                   | 0.735       |
|                                                                                                 | ENV14         | AI automatically flags risk points in its output content                                                                       | 0.743       |
|                                                                                                 | Reverse ENV2  | AI produces noticeably different outputs when executing the same instructions repeatedly                                       | 0.706       |
|                                                                                                 | ENV3          | AI lacks transparency in the process of result generation                                                                      | 0.765       |
|                                                                                                 | ENV4          | AI prevents the specific sources of potential problems in its outputs from being identified                                    | 0.759       |
|                                                                                                 | ENV5          | AI conceals potential moral risks during interactions                                                                          | 0.711       |
|                                                                                                 | ENV6          | AI provides a detailed explanation of its content generation process on the user interface                                     | 0.723       |
|                                                                                                 | Reverse ENV7  | AI exhibits randomness and uncertainty when generating content                                                                 | 0.748       |
|                                                                                                 | ENV9          | AI makes updates and changes within the system invisible to users                                                              | 0.697       |
|                                                                                                 | ENV10         | AI provides users with aids to help understand the interaction process                                                         | 0.667       |
|                                                                                                 | Reverse COG1  | When using AI, I find it difficult to clearly judge how much moral responsibility I should bear                                | 0.856       |
|                                                                                                 | COG11         | The longer I use AI tools, the less clear my recognition of my own moral responsibility becomes                                | 0.735       |
|                                                                                                 | COG12         | When facing ambiguous AI outputs, I find it difficult to distinguish the boundary of responsibility between myself and AI      | 0.782       |
| Ambiguity in Responsibility Attribution (Cronbach's $\alpha = 0.953$ , CR = 0.954, AVE = 0.654) | COG2          | If I fail to detect moral issues in AI content in time, I am uncertain whether I should be held responsible afterwards         | 0.836       |
|                                                                                                 | COG3          | For hidden moral issues in AI content (such as implicit bias), I cannot clearly determine who is at fault                      | 0.838       |
|                                                                                                 | COG4          | When facing the results of moral risks generated in AI interaction, I feel that responsibility can be clearly defined          | 0.840       |
|                                                                                                 | Reverse COG5  | When faced with unethical conclusions from AI, I find it difficult to generate a clear sense of responsibility                 | 0.813       |
|                                                                                                 | COG7          | When AI generates ambiguous information, I feel the attribution of responsibility is clear                                     | 0.807       |
|                                                                                                 | Reverse COG8  | When using AI, I feel my sense of moral responsibility gradually becomes blurred                                               | 0.757       |
|                                                                                                 | COG9          | I usually cannot judge whether ambiguous information generated by AI deserves to be handled responsibly by me                  | 0.820       |
|                                                                                                 | COG10         | When facing controversial content generated by AI, I doubt the extent to which I should bear moral responsibility              | 0.804       |
|                                                                                                 | BEH1          | When AI output seems questionable, action can wait because the system will correct itself                                      | 0.678       |
|                                                                                                 | BEH10         | I tend to trust that the AI is designed ethically, so I don't feel the need to monitor it                                      | 0.804       |
| Passive Responsibility Exemption (Cronbach's $\alpha = 0.960$ , CR = 0.960, AVE = 0.635)        | BEH11         | I see it as beyond my duty to give feedback or intervene when I spot ethical risks in AI content                               | 0.829       |
|                                                                                                 | BEH12         | I tend to trust the AI's default settings to handle ethical issues, so I skip additional checks                                | 0.841       |
|                                                                                                 | BEH13         | In uncertain situations, I usually don't take action to prevent ethical problems with AI                                       | 0.797       |
|                                                                                                 | BEH14         | I consider the potential ethical risks posed by AI to be serious                                                               | 0.755       |
|                                                                                                 | Reverse BEH2  | I don't usually make an effort to verify the ethical risks of AI-generated information                                         | 0.827       |
|                                                                                                 | BEH3          | When AI content has ethical controversies, I don't feel the need to adjust or correct it                                       | 0.758       |
|                                                                                                 | BEH4          | If errors arise from AI suggestions, I consider myself responsible for the outcome rather than the tool                        | 0.768       |
|                                                                                                 | Reverse BEH5  | I believe that users bear the responsibility for addressing AI's ethical problems, rather than leaving it solely to developers | 0.760       |
|                                                                                                 | Reverse BEH6  | I trust the platform to handle any serious ethical issues, so I don't need to be concerned                                     | 0.787       |
|                                                                                                 | BEH7          | I rarely think about what ethical principles the AI follows when I use it                                                      | 0.846       |
|                                                                                                 | BEH8          | I don't typically reflect deeply on the potential ethical risks in AI content                                                  | 0.857       |
|                                                                                                 | BEH9          | When AI output is ethically unclear, I usually accept it without questioning                                                   | 0.825       |

**Table 9.** The discriminant validity of the MSS.

| Dimensions                              | Factor1 | Factor2 | Factor3 |
|-----------------------------------------|---------|---------|---------|
| Interactive Environmental Complexity    | 0.727   |         |         |
| Ambiguity in Responsibility Attribution | 0.639   | 0.809   |         |
| Passive Responsibility Exemption        | 0.088   | 0.209   | 0.797   |

**Table 10.** Chi-squared difference test.

|                | Df   | AIC   | BIC   | Chisq    | Chisq diff | RMSEA | Df diff | P     |
|----------------|------|-------|-------|----------|------------|-------|---------|-------|
| fit_configural | 1252 | 32241 | 33128 | 1980.500 |            |       |         |       |
| fit_metric     | 1286 | 32204 | 32960 | 2012.400 | 31.915     | 0.000 | 34      | 0.570 |
| fit_scalar     | 1320 | 32180 | 32803 | 2056.200 | 43.799     | 0.040 | 34      | 0.121 |
| fit_strict     | 1357 | 32148 | 32627 | 2097.800 | 41.572     | 0.026 | 37      | 0.278 |

Harman's single-factor test showed that the first factor accounted for 33.301% of the variance, confirming that common method bias in this dataset was within acceptable limits.

### 3.6. Phase V: Measurement invariance and finalization of the scale

Building upon the results of the CFA, this study further assessed the measurement invariance of the Moral Suspension Scale across gender groups, ensuring both the theoretical basis and empirical validity for cross-group comparisons (van de Schoot et al., 2012). The measurement invariance analysis employed a multi-group confirmatory factor analysis (MG-CFA) with a hierarchical, stepwise approach, sequentially testing for configural invariance, metric invariance (factor loading invariance), scalar invariance (intercept invariance), and strict invariance (residual invariance).

The criteria for establishing measurement invariance followed international standards (Putnick & Bornstein, 2016), requiring CFI and TLI values above 0.90, changes in  $\Delta$ CFI and  $\Delta$ TLI less than 0.01, RMSEA less than 0.08 with  $\Delta$ RMSEA less than 0.015, and SRMR less than 0.08 with  $\Delta$ SRMR less than 0.03.

The results are shown in Tables 10–12. First, the configural invariance model results showed  $\chi^2 = 1980.528$ ,  $df = 1252$ ,  $RMSEA = 0.057$ ,  $CFI = 0.929$ ,  $TLI = 0.924$ , and  $SRMR = 0.054$ , indicating good model fit and demonstrating that the three-factor structure was adequately reproduced across gender groups. Imposing equality constraints on factor loadings resulted in the metric invariance model ( $\chi^2 = 2012.443$ ,  $df = 1286$ ,  $RMSEA = 0.056$ ,  $CFI = 0.929$ ,  $TLI = 0.927$ ,  $SRMR = 0.058$ ). Comparisons between the metric and configural models yielded  $\Delta$ CFI = 0.000,  $\Delta$ RMSEA =  $-0.001$ , and  $\Delta$ SRMR = 0.004, with the chi-square difference test ( $\Delta\chi^2 = 31.915$ ,  $\Delta df = 34$ ,  $p = 0.570$ ) being non-significant, indicating that factor loadings were equivalent across gender, thus meeting the criteria for metric invariance.

Further, by constraining the intercepts of the observed variables, the scalar invariance model was obtained ( $\chi^2 = 2056.242$ ,  $df = 1320$ ,  $RMSEA = 0.055$ ,  $CFI = 0.928$ ,  $TLI = 0.928$ ,  $SRMR = 0.059$ ). Compared to the metric invariance model,  $\Delta$ CFI =  $-0.001$ ,  $\Delta$ RMSEA = 0.000,  $\Delta$ SRMR = 0.000, with the chi-square difference test ( $\Delta\chi^2 = 43.799$ ,  $\Delta df = 34$ ,  $p = 0.121$ ) being statistically non-significant. The strict invariance model, which additionally constrained residual variances, yielded  $\chi^2 = 2097.814$ ,  $df = 1357$ ,  $RMSEA = 0.055$ ,  $CFI = 0.928$ ,  $TLI = 0.929$ ,  $SRMR = 0.058$ . Compared to the scalar model,  $\Delta$ CFI = 0.000,  $\Delta$ RMSEA =  $-0.001$ ,  $\Delta$ SRMR = 0.000, with the chi-square difference test ( $\Delta\chi^2 = 41.572$ ,  $\Delta df = 37$ ,  $p = 0.278$ ) also being non-significant. All fit indices satisfied the empirical criteria for measurement invariance (Putnick & Bornstein, 2016), with changes in CFI, RMSEA, and SRMR well below the critical thresholds. Therefore, it can be concluded that the scale achieved acceptable levels of configural, metric, scalar, and strict invariance across gender groups, providing a strong basis for cross-group comparison.

Taking into account the results from the preceding exploratory and confirmatory factor analyses, reliability and validity tests, as well as the measurement invariance assessment by gender, this study ultimately finalized the Moral Suspension Scale as comprising three dimensions: Interactive Environmental Complexity, Ambiguity in Responsibility Attribution, and Passive Responsibility Exemption, with a total of 38 items. The scale demonstrated robust structural validity and parameter

**Table 11.** Model fit indices.

|                | Chisq    | Df   | P     | RMSEA | CFI   | TLI   | SRMR  | AIC       | BIC       |
|----------------|----------|------|-------|-------|-------|-------|-------|-----------|-----------|
| fit_configural | 1980.528 | 1252 | 0.000 | 0.057 | 0.929 | 0.924 | 0.054 | 32240.558 | 33128.482 |
| fit_metric     | 2012.443 | 1286 | 0.000 | 0.056 | 0.929 | 0.927 | 0.058 | 32204.473 | 32959.987 |
| fit_scalar     | 2056.242 | 1320 | 0.000 | 0.055 | 0.928 | 0.928 | 0.059 | 32180.272 | 32803.377 |
| fit_strict     | 2097.814 | 1357 | 0.000 | 0.055 | 0.928 | 0.929 | 0.058 | 32147.844 | 32626.856 |

**Table 12.** Differences in fit indices.

|                           | Df | $\Delta$ RMSEA | $\Delta$ CFI | $\Delta$ TLI | $\Delta$ SRMR | AIC     | BIC      |
|---------------------------|----|----------------|--------------|--------------|---------------|---------|----------|
| fit_metric-fit_configural | 34 | −0.001         | 0.000        | 0.002        | 0.004         | −36.085 | −168.495 |
| fit_scalar-fit_metric     | 34 | 0.000          | −0.001       | 0.001        | 0.000         | −24.201 | −156.610 |
| fit_strict-fit_scalar     | 37 | −0.001         | 0.000        | 0.002        | 0.000         | −32.428 | −176.521 |

stability across gender groups, making it a reliable tool for subsequent cross-group comparisons and applied research, and providing a solid foundation for measuring and exploring the mechanisms of moral suspension in both academic and professional contexts.

## 4. Discussion

While the widespread adoption of GenAI brings significant benefits, it also raises significant ethical concerns. Prior research on GenAI evaluation has mainly focused on aspects such as generation quality, scalability, stability, interpretability, and collaborative capabilities (Liu et al., 2023; Morabito & Jang, 2025; Perifanou & Economides, 2025). In contrast, our study centers on the morally ambiguous issue faced by Chinese emerging adults in human-GenAI interaction contexts (specifically regarding academic and creative tasks), and innovatively introduces the theoretical construct of Moral Suspension. It is important to note that the Moral Suspension Scale was developed and validated specifically using data from Chinese emerging adults who utilized GenAI tools to complete academic assignments or creative projects. Consequently, the scale is most appropriately applied to populations and contexts aligning with these characteristics.

Moreover, our study provides a new measurement tool for assessing moral ambiguity in GenAI contexts, and brings to light a series of theoretical and practical insights that are worth further reflection. By clarifying the psychological mechanisms underlying moral suspension, our results invite a reexamination of existing theories of moral responsibility in technologically mediated environments. At the same time, the scale developed here offers a concrete basis for future research, educational intervention, and system design, making it possible to respond more effectively to the ethical complexities now emerging in digital society. In this section, we further elaborate on the theoretical contributions and practical significance of our work, and consider its implications for both research and practice.

### 4.1. Theoretical contributions

The core theoretical contribution of our study lies in proposing and operationalizing the new concept of moral suspension, which opens up a novel theoretical pathway for understanding the moral psychological mechanisms underpinning human-GenAI collaboration. Its theoretical significance is reflected in three main aspects.

First, the introduction of moral suspension facilitates a deeper understanding of the complexity of moral cognition in the era of AI and enables constructive dialogue with traditional theories. Previous studies have predominantly focused on constructs such as moral disengagement, emphasizing individuals' deliberate cognitive strategies to evade responsibility (Bandura, 1999a; Newman et al., 2020). However, such frameworks fall short in explaining the “involuntary” ethical confusion users face when interacting with opaque and autonomous systems like GenAI. The construct of moral suspension captures this state of “neither fully responsible nor entirely exempt” that emerges passively due to technological constraints (e.g., algorithmic opacity) and cognitive limitations (e.g., difficulty in discerning individual contribution) (Malle, 2016; Waqas et al., 2025). Unlike moral disengagement, it does not

involve conscious avoidance of moral responsibility, but rather a passive cognitive ambiguity triggered by the pressure of external technological environments.

Second, this study offers a micro-level psychological explanation for macro-level ethical debates such as the responsibility gap. The concept, proposed by Matthias (2004), refers to the challenge of attributing accountability for the outcomes of autonomous machine behavior to any specific agent. By developing a three-dimensional structure of moral suspension, this study provides a nuanced account of how such gaps may emerge from the interactive environment and users' psychological responses. In this sense, the responsibility gap is not merely a legal or institutional issue but also reflects the real cognitive pressure and attributional ambiguity experienced by users during human-GenAI interaction. In particular, the dimension of Interactive Environmental Complexity directly addresses why explainability (XAI) is crucial (Arrieta et al., 2020; Ross, 2024; F. Xu et al., 2019). An opaque AI system fundamentally obstructs users from forming clear moral judgments, thereby creating psychological conditions conducive to moral suspension.

Third, the theoretical model developed in this study extends the application of Social Cognitive Theory to the context of AI. According to Bandura's (1989) triadic reciprocal model, individual behavior results from the ongoing mutual construction among personal, behavioral, and environmental factors. The three dimensions of moral suspension align closely with this model: Ambiguity in Responsibility Attribution reflects cognitive-level confusion; Passive Responsibility Exemption captures the weakening of responsibility at the behavioral level; and Interactive Environmental Complexity underscores the role of GenAI as an environmental determinant. This framework transcends the binary view of AI as either a "tool" or an "actor" (Sundar, 2020), instead conceptualizing it as a dynamic environmental force that shapes users' moral cognition. It thus provides an integrative analytical framework for future research in human-GenAI interaction, enabling systematic examination of the interplay between technological features, individual psychology, and ethical behavior. This is particularly valuable in the context of emerging adults, a demographic in the formative stage of value development, offering a theoretical foundation for understanding how technology may shape the moral frameworks of the next generation.

#### 4.2. Practical implications

In addition to its theoretical contributions, the Moral Suspension Scale developed in this study holds considerable practical potential. It can support responsible AI system design, educational interventions, and ethical policy development.

For AI developers and designers, the Moral Suspension Scale serves as a diagnostic and evaluative instrument for promoting Responsible AI. Current ethical design principles for AI often remain at the abstract level, emphasizing notions such as transparency and explainability (Fernandez-Quilez, 2023; Kim et al., 2020). By contrast, the scale offers a set of quantifiable indicators reflecting users' psychological experiences. Developers can use the scale to evaluate how different design elements, such as user interfaces, explanatory prompts, or interaction modalities, affect users' states of moral suspension throughout iterative product development. This enables developers not only to fulfill their social responsibilities more effectively but also to enhance user trust and product competitiveness, translating abstract ethical principles into concrete design practices.

For educators and institutional administrators, the Moral Suspension Scale provides a scientific foundation for implementing digital ethics education and targeted interventions. Emerging adults are not only the core users of GenAI but also key audiences for academic integrity and professional ethics training (Birks & Clare, 2023). Educators and managers can apply the scale to identify students or employees who exhibit high levels of moral suspension in AI usage contexts, and then tailor instructional content and training programs accordingly. This shifts academic integrity education from the traditional emphasis on "plagiarism prevention" to a more profound goal of cultivating "responsible users in the AI era." Such efforts help students and staff enjoy the affordances of AI while simultaneously developing clear moral boundaries and critical thinking skills to respond to emerging challenges like AIgicism (Tang, 2024).

For policymakers and regulatory agencies, this study provides empirical insights for refining AI governance frameworks. Post hoc accountability is insufficient; proactive risk prevention is imperative. Policies can guide or mandate the integration of design elements into GenAI tools, especially in education and research domains, that aim to mitigate moral suspension. For instance, requirements might include compelling AI systems to explain their content-generation logic in clear, non-technical language or to include a “responsibility acknowledgment” step before users export content, reminding them to verify information independently and assume final responsibility. The Moral Suspension Scale can be employed to assess the effectiveness of such ethics-embedded design initiatives, thereby promoting a shift in industry standards from mere functionality toward a dual emphasis on psychological well-being and ethical guidance.

#### **4.3. Limitations and future work**

Although this study employed a systematic, stepwise approach to develop and validate the Moral Suspension Scale, several limitations must be acknowledged. First, the sample consisted exclusively of emerging adults in China who utilized GenAI tools primarily for academic or creative tasks. Specifically, all participants were native Chinese speakers, and the survey was administered in Chinese. While the English translations of the items are provided in this paper for reporting purposes, the psychometric properties were established based on the Chinese version. While this sampling strategy aligns closely with the actual user demographic of GenAI, it may limit the generalizability of the scale across different national, cultural, and linguistic contexts. Therefore, caution is warranted when interpreting results if the scale is administered in English or other non-Chinese languages, and further cross-linguistic validation is recommended.

Second, the data collection method was a self-reported questionnaire survey. Although various techniques were employed, such as response time controls, reverse-scored items, and trap items, to mitigate subjective bias, the results may still be influenced by social desirability effects. To improve validity, future studies are encouraged to integrate objective data sources such as behavioral experiments, human-AI interaction logs, or digital trace analysis, which would enable a more comprehensive understanding of the relationship between moral suspension and observable behaviors.

Third, as AI systems become increasingly autonomous, personified, and socially embedded, the manifestations and mechanisms of moral suspension may evolve. Future work could incorporate additional contextual variables such as AI capability perception, institutional norms, and ethical climate to refine the contextual specificity and dynamic adaptability of the measurement tool. Moreover, to establish the short- and long-term stability of the scale, its test-retest reliability could be assessed in longitudinal designs. This would enhance the scale’s utility as a tool for monitoring users’ evolving moral cognition and for evaluating the effectiveness of ethics education interventions over time. By continuously expanding both its applied contexts and theoretical depth, the Moral Suspension Scale has the potential to become a foundational instrument for AI ethics governance and the promotion of digital psychological well-being in the emerging socio-technical landscape.

### **5. Conclusions**

In this paper, our study centers on the morally ambiguous issue faced by Chinese emerging adults in human-GenAI interaction contexts, and innovatively introduces the theoretical construct of Moral Suspension. Through a rigorous scale development procedure, we constructed and validated the Moral Suspension Scale to address the need for a rigorous and practical tool for measuring ethical ambiguity in GenAI-assisted scenarios. The development process was anchored in the triadic reciprocal model of social cognitive theory, with the conceptualization of moral suspension grounded in recent interdisciplinary research. Item generation drew upon both theoretical considerations and the actual uncertainties encountered by users in technology-mediated environments. After multiple rounds of expert review and pilot testing, the instrument was refined to ensure both content validity and contextual relevance. The overall scale and its three dimensions demonstrated internal consistency reliability above 0.877, and exhibited strong psychometric properties, including convergent and discriminant validity as well as measurement invariance across gender groups. Ultimately, this study developed a three-dimensional scale consisting of

38 items, encompassing the three dimensions. The scale provides a reliable instrument for quantitatively assessing users' passive moral cognition uncertainties in GenAI-enabled scenarios.

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## Author contributions

CRedit: **Zhirui Chen**: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft; **Wenchen Guo**: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft; **Yiyang Zhang**: Data curation, Formal analysis, Writing – original draft; **Xihui Liu**: Funding acquisition, Supervision, Writing – original draft; **Biao Li**: Project administration, Supervision; **Hailiang Wang**: Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

## Ethical approval

The study has been approved by the Institutional Review Board of The Hong Kong Polytechnic University [Reference number: HSEARS20250714003]. All participants read an informed consent statement and agreed to participate voluntarily before completing the online survey.

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